

Goal

Estimate how long it takes to create `acc_mv` under approach 1.

Problem

`acc_mv` is not loaded directly from disk unless you saved it separately. It is created in two steps:

```
acc_long <- load_metadata_table("acc_long")
acc_mv    <- make_acc_mv_from_long(acc_long)
```

So the time includes:

1. reading all `acc_long` partition files from `metadata_output/acc_long_parts/` ;
2. binding the partitions together;
3. reshaping/summarizing `acc_long` into `acc_mv` .

Practical estimate

With **81 chunks** from `HAKE_METADATA_CHUNK_N = "12"` :

Case	Likely time to create <code>acc_mv</code>
Fast SSD, <code>data.table</code> installed, moderate file size	1–5 minutes
Standard Windows laptop, compressed RDS files, large <code>acc_long</code>	5–15 minutes
Very large metadata tables or slow network/OneDrive folder	15–45+ minutes

The biggest time cost will probably be reading and binding `acc_long` , not the final `make_acc_mv_from_long()` step.

Exact timing syntax

Run this in a fresh R session:

```
rm(list = ls())
gc()

setwd("C:/Users/Jon.Brodziak/Desktop/Dirichlet Study/hake/RUN1")

Sys.setenv(HAKE_RUN_ON_SOURCE = "false")
Sys.setenv(HAKE_METADATA_OUTDIR = file.path(getwd(), "metadata_output"))

source("metadata.R")
```

```
t_load_acc_long <- system.time({
  acc_long <- load_metadata_table("acc_long")
})

print(t_load_acc_long)

format(object.size(acc_long), units = "GB")
nrow(acc_long)
```

Then time the creation of `acc_mv` :

```
t_make_acc_mv <- system.time({
  acc_mv <- make_acc_mv_from_long(acc_long)
})

print(t_make_acc_mv)

format(object.size(acc_mv), units = "GB")
nrow(acc_mv)
```

Then report total time:

```
t_load_acc_long + t_make_acc_mv
```

Better diagnostic before loading everything

First check the compressed size of the partition files:

```
acc_long_files <- list.files(
  "metadata_output/acc_long_parts",
  pattern = "^acc_long_part_[0-9]+\\.rds$",
  full.names = TRUE
)

length(acc_long_files)

sum(file.info(acc_long_files)$size) / 1024^3
```

Rule of thumb:

```
compressed_size_gb <- sum(file.info(acc_long_files)$size) / 1024^3
```

Approximate runtime:

Compressed acc_long size	Expected load + acc_mv time
< 0.5 GB	usually under 2 minutes
0.5–2 GB	about 2–10 minutes
2–5 GB	about 10–30 minutes
> 5 GB	may take 30+ minutes and may hit RAM limits

Time a small subset first

This estimates runtime without loading all 81 chunks:

```
rm(list = ls())
gc()

setwd("C:/Users/Jon.Brodziak/Desktop/Dirichlet Study/hake/RUN1")

Sys.setenv(HAKE_RUN_ON_SOURCE = "false")
Sys.setenv(HAKE_METADATA_OUTDIR = file.path(getwd(), "metadata_output"))

source("metadata.R")

t_test <- system.time({
  acc_long_test <- load_metadata_table("acc_long", part_ids = 1:5)
  acc_mv_test <- make_acc_mv_from_long(acc_long_test)
})

print(t_test)

nrow(acc_long_test)
nrow(acc_mv_test)

rm(acc_long_test, acc_mv_test)
gc()
```

Approximate the full time:

```
estimated_full_seconds <- as.numeric(t_test["elapsed"]) * 81 / 5
estimated_full_minutes <- estimated_full_seconds / 60

estimated_full_minutes
```

Best practical recommendation

Install `data.table` before loading:

```
install.packages("data.table")
```

Then run:

```
acc_long <- load_metadata_table("acc_long")
acc_mv    <- make_acc_mv_from_long(acc_long)
```

The revised loader uses `data.table::rbindlist()` automatically when `data.table` is available, which should reduce the time and memory overhead of combining the 81 partitions.